MONTH 1: FRONT-END DEVELOPMENT

- Week 1: HTML & CSS
- Tasks:
  - Learn the basics of HTML5 and CSS3.
  - Create a static webpage with forms, tables, and basic styling.
- Project: Build a personal portfolio website with multiple pages.
- Week 2: Responsive Design
- Tasks:
  - Learn responsive design using CSS Flexbox and Grid.
  - Implement media queries for different screen sizes.
- Project: Convert the portfolio website into a fully responsive website.
- Week 3: JavaScript Basics
- Tasks:
  - Understand JavaScript syntax, DOM manipulation, and events.
  - Learn basic debugging and testing techniques.
- Project: Add interactivity to the portfolio website (e.g., a dynamic image gallery or a contact form with validation).
- Week 4: JavaScript Frameworks (React.js)
- Tasks:
  - Learn the basics of React.js.
  - Understand components, state, and props.
- Project: Develop a simple to-do list application using React.js.

MONTH 2: BACK-END DEVELOPMENT

- Week 5: Introduction to Back-End (Node.js/Express)
- Tasks:
  - Set up a basic server using Node.js and Express.
  - Learn about RESTful APIs and routing.
- Project: Create a simple RESTful API for managing to-do items.
- Week 6: Database Integration (MongoDB)
- Tasks:
  - Learn MongoDB basics and Mongoose ORM.
  - Implement CRUD operations for the to-do API.
- Project: Integrate MongoDB with the to-do API and add user authentication.
- Week 7: Authentication & Authorization
- Tasks:
  - Implement user authentication using JWT or OAuth.
  - Secure the API with role-based access control.
- Project: Add secure user login/logout functionality to the to-do application.
- Week 8: Full-Stack Project
- Project: Build a full-stack application combining front-end (React) and back-end (Node.js, MongoDB) to create a small web app, such as a blog platform or e-commerce site.

MONTH 3: ADVANCED TOPICS AND DEPLOYMENT

- Week 9: Version Control & Collaboration
- Tasks:
  - Learn Git and GitHub for version control.
  - Understand branching, merging, and pull requests.
- Project: Collaborate with peers on a group project using GitHub.
- Week 10: Deployment (Heroku, Netlify)
- Tasks:
  - Learn to deploy web applications using platforms like Heroku and Netlify.
- Project: Deploy the full-stack application developed in Week 8.
- Week 11: Performance Optimization
- Tasks:
  - Learn about optimizing web performance (e.g., caching, minification, lazy loading).
- Project: Optimize the deployed application and measure improvements.
- Week 12: Final Project & Presentation
- Project: Choose a final project (e.g., social media platform, task management system) and develop it from scratch. Present the project with a live demo and documentation.